



Guidelines for Ethics by Design

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Executive summary

This present deliverable builds on the findings of deliverable 5.1 (state of the art and literature review) and also informs deliverable 5.1 (survey and interview design). The objective of this deliverable is to provide guidelines and tools for the operationalisation of “ethics-by-design” methodology within the GuestXR project. The guidelines and tools contained in the deliverable are intended to be used by members of the consortium or associated projects together with the participation of ethics and responsible innovation researchers. However, researchers in XR and artificial intelligence can also use insights and resources from these guidelines independently. The deliverable begins (section 2) by sketching several relevant approaches that fall under the broader umbrella term of ethics-by-design. These include value-sensitive design and Virtual Ethnography and Future Studies. The deliverable then provides a toolkit (section 3) for researchers to use in the implementation of ethics-by-design approaches within design and innovation processes. This toolkit is best implemented in the context of close collaboration with ethics and/or responsible innovation researchers, for example in the context of a research stay or “embedding” of an ethics and/or responsible innovation researcher within a laboratory or relevant scientific group. The toolkit is divided into four potential stages of implementation in the ethics-by-design approach: Prepare, Explore, Understand, Improve & Implement. Each stage within section 3 then contains several tools that can be utilised in the implementation of the ethics-by-design approach.

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List of abbreviations

Abbreviation	Explanation
AI	Artificial Intelligence
AOIR	Association of Internet Researchers
EC	European Commission
GDPR	General Data Protection Regulation
HCI	Human-Computer Interaction
ICE	Information and Computing Ethics
IVE	Intelligent Virtual Environment
KPI	Key Performance Indicator
ML	Machine Learning
MM	Midstream Modulation
R&D	Research and Development
STIR	Socio-Technical Integration Research
UM	Maastricht University
VR	Virtual Reality
VSD	Value Sensitive Design

Introduction

Today, ethics is often used as an umbrella term for addressing normative concerns across a variety of domains and therefore typically encompasses more than what we might imagine as traditional philosophical ethical questions. For example, in the domains of artificial intelligence (AI) and advanced computing, ethics often includes questions about compliance with existing legislation (e.g. GDPR), as well as questions about governance, and broader societal impacts.

Perhaps as a result of this lack of specificity, ethics sometimes risks being seen merely as a checklist activity or an activity in legal compliance, and whilst this is important (there is after all, a time and a place for checklists) we want ethics to mean more than that. We want to avoid simply framing ethics as some new form of self-regulation, and likewise avoid a situation where ethics becomes seen as an “easy” or “soft” alternative to legal or regulatory frameworks (Wagner 2018).

Ethics by Design is not:

-  A checklist activity
-  A form of legal compliance
-  A means of gate keeping
-  A KPI
-  An additional task

The approaches and tools outlined throughout these guidelines fall under the broad banners of **ethics by design**, **human-centered design** and **responsible innovation**. Essentially, what these approaches share is the objective of making research and innovation activities more responsive to ethical concerns and questions of values; as well as making ethical reflection and subsequent initiatives more accessible to researchers, practitioners, users and other stakeholders.

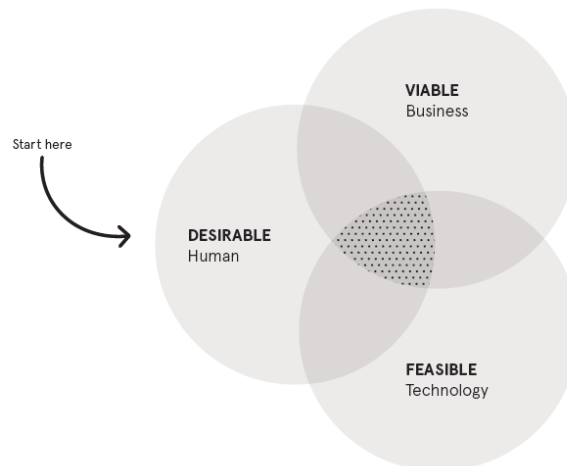


Figure 1: Human-Centered Design
(from *The Field Guide to Human-Centered Design*, 2015).

What makes ethics by design approaches different from other user-centred design methodologies is that they typically involve asking ethical questions right from the very start, involving diverse stakeholders in ongoing discussions, so as to embed values of ethical importance throughout the design and development process. These approaches then aim to steer technological artefacts towards desirable goals, by incorporating the needs and values of users, thereby presumably improving their moral and societal acceptability in the long run.

Ethics by Design is:

- ☒ Part of design and development processes
- ☒ A form of problem solving
- ☒ A way of navigating tricky issues
- ☒ A means of evaluating trade-offs

While few people would argue against efforts aimed at making research and innovation more responsible, putting ethics into practice is not easy. Certainly, when it comes to AI, information about how to translate abstract ethical values into the actual work of design typically goes unspecified (for comprehensive reviews see Attard-Frost, De los Ríos & Walters, 2022; Hagendorff, 2020; and Jobin, Ienca & Vayena, 2019). This is perhaps because, as Donald McMillan and Barry Brown (2019) acknowledge

Design requires tradeoffs and, in practice, it can be difficult to see how to balance conflicting ethical principles, or even how to respect these principles without producing designs that are poorer in some critical aspect (p. 2).

The concept of Responsible Innovation (RI) functions as a kind of backbone to methodologies grouped under the umbrella of ethics by design. Responsible Innovation is based on four basic principles: anticipation, reflexivity, inclusion, and responsiveness (Stilgoe et al, 2013). These principles provide a framework for researchers and innovators, encouraging them to consider possible unintended consequences (anticipation), their underlying assumptions and motivations (reflexivity), the role of relevant stakeholders (inclusion), and the ways in which new knowledge is produced as a result of these questions (responsiveness).

Principles of RI	Description
Anticipation	<i>Anticipation is about carefully examining both the intended and possible unintended consequences arising from research and innovation activities, including environmental, health-related, economic and social impacts. Anticipatory processes prompt “what if...?” questions that allow researchers and innovators to pre-prepare for and respond to the various uncertainties and dilemmas built into their work.</i>
Reflexivity	<i>Reflexivity is about reflecting on the underlying motivations, assumptions and commitments driving research and innovation. It commits researchers and innovators to inquire and challenge the taken-for-granted assumptions structuring their work and makes them attentive to alternative ways of framing the value and societal impact of their ideas, methods and proposed solutions.</i>
Inclusion	<i>Inclusion is closely related to public engagement and stakeholder involvement. It is about involving relevant societal actors in research and innovation activities from an early stage, and ensuring continuous, open dialogue about desirable and undesirable outcomes throughout the project. Inclusion serves to broaden the ideas, perspectives and world-views guiding research and innovation activities.</i>
Responsiveness	<i>Responsiveness is about aligning research and innovation activities with the new perspectives, insights and values emerging through anticipatory, reflexive and inclusion-based RRI processes. Responsiveness presupposes a will to learn from practical experience and a capacity to translate this learning into better, more responsible research and innovation solutions.</i>

Table 1: Principles of Responsible Innovation



However, translating these principles into practice is still easier said than done (Blok and Lemmens 2015; Burget et al. 2017, Felt, 2017). The purpose of these guidelines is therefore to provide a detailed overview of ethics by design approaches, as well as a number of practical tools and methods for implementing responsible innovation in practice.

1.1 How to use these guidelines

In these guidelines we have gathered various resources, which all provide different entry points for thinking about ethics by design. These resources have been drawn from our own experiences working with ethics by design approaches, as well as from the output of other European Commission (EC) funded research projects. The guidelines are also informed by the literature review carried out as part of WP1 (D5.1).

However, despite the tools outlined in this document being based on different disciplinary perspectives, philosophies, and methodologies, they are all based on the same two principles, that technologies are not value neutral and that ethics is therefore a necessary part of the design process.



Not a Recipe

These guidelines suggest a number of different tools for doing ethics by design, not a clear-cut recipe.

Key Principles

Technologies are not value neutral

&

Ethics is a part of the design process

In the following sections, we outline a number of approaches which fall under the banner of ethics by design, before providing an overview of specific tools that can be deployed at various stages throughout the project. You can of course read the guidelines from cover to cover, but you can also navigate to the sections that are most relevant to you at any given moment in time. Essentially, these guidelines are intended to provide an overview of the types of activities the team from Maastricht University (UM) will utilize to guide partners through how to embed ethics by design throughout the GuestXR project.

It is important to keep in mind that these guidelines are an evolving document and as such consortium members are encouraged to revisit them regularly as they iterate between various phases of the GuestXR project life-cycle.



Building upon the NewHoRRizon project's *Thinking Tool* (2018) for researchers and innovators who wish to reflect on the societal readiness of their projects, the organization of the guidelines is inspired by a stage-gating approach. A commonly used business strategy in product development, stage-gating is based upon a linear approach to the development cycle.

R&D Phase	Description
Research Design & Problem Formulation	<i>Captures the ideation process, where new ideas for discovery are conceptualized, research problems are formulated and appropriate procedures for data collection and experimentation are planned.</i>
Data Collection	<i>Covers activities related to data collection.</i>
Data Analysis & Evaluation	<i>Encompasses data analysis, evaluation and interpretation of results.</i>
Prototyping, Testing & Launching	<i>Covers the phase within which prototypes are developed and tested, improvements are made, and the product or service is launched and/or results are disseminated to relevant stakeholders, researchers and public audiences</i>

Table 2: Description of Common Research Phases

Stage-gating identifies different phases as distinct and punctuated by so-called decision gates. From a business perspective, the assessment criteria used to assess whether or not a product can progress to the next stage, are typically aimed at increasing cost-efficiency and driving market value.

From the perspective of ethics by design however, these decision gates can instead be seen as reflection gates, providing key moments for reflection on the extent to which adequate ethical reflection has been embedded throughout each phase of the project's life-cycle.

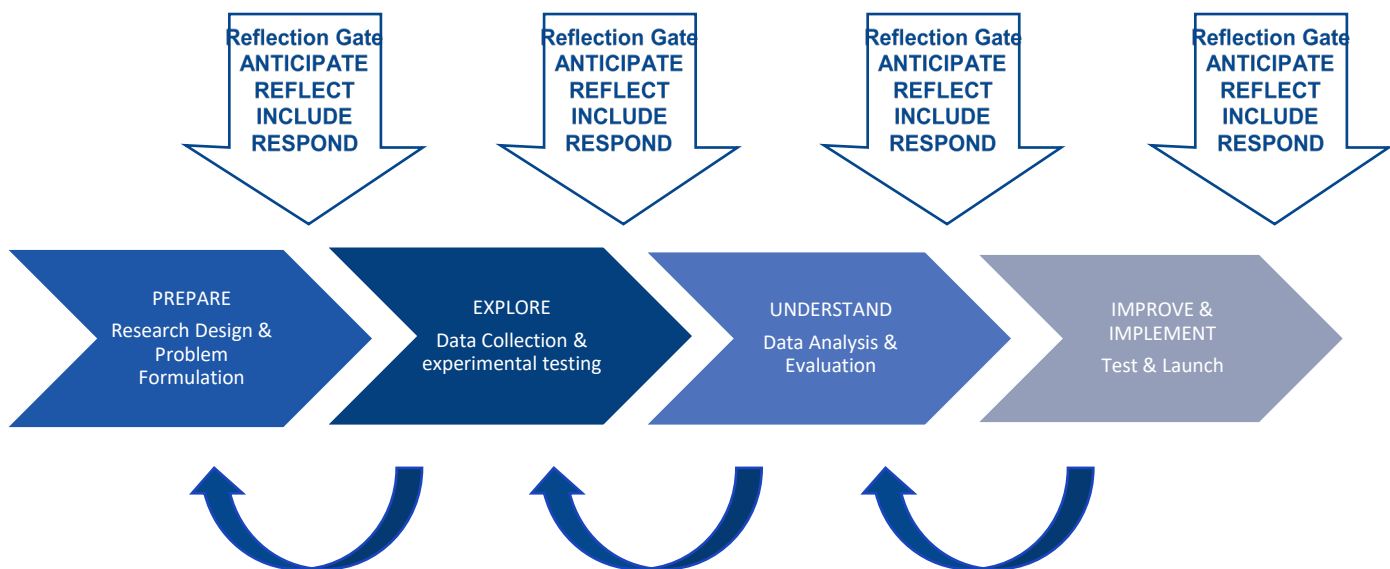


Figure 1: Iterative Phases of Innovation and Reflection Gates

At different reflection gates, different approaches and methods could be used in order to ensure that ethical reflection is embedded throughout the innovation process (see Figure 1). While stage-gating provides a useful means for ensuring that moments of reflection are built into research and design (R&D) practices, it is **crucial** to acknowledge that R&D processes are **rarely, if ever, linear**. Methods such as Service Design stress the iterative nature of design processes in practice, outlining steps from preparation (defining the challenges) and exploration (identifying the needs), to understanding the needs, improving, and implementing the solutions. As demonstrated in figure 1, these steps compliment the phases outlined by the stage-gating approach, whilst also emphasising the extent to which designers and developers go back and forth based on the lessons learned and knowledge gained along the way. For example, feedback loops between phases are common and in large consortiums, project members typically work on different phases in parallel (NewHoRRizon, 2018).

In the sections that follow, we will outline tools and methods that can be used across various phases, as well as questions which can serve as prompts for evaluation at each of the reflection gates. There are a broad range of tools that incorporate ethics by design elements and can be used in co-creation processes to facilitate engagement between designers, developers, and users and foster more responsible practices. These tools serve the purposes of finding solutions, structuring information, collecting ideas, assessing impacts, fostering empathy, or learning about needs (Living Innovation, 2019).



2 Ethics by Design Approaches

As outlined by the Association of Internet Researchers (AoIR), there are a range of ethical frameworks which can be applied in order to try and resolve ethical dilemmas and challenges. They suggest that deontology and utilitarianism, feminist ethics, ethics of care, and virtue ethics have become increasingly common within Information and Computing Ethics (ICE). In their Internet Research: Ethical Guidelines 3.0, the AoIR endorses ethical pluralism. They explain that “the issues raised by Internet research are *ethical* problems precisely because they evoke more than one ethically defensible response to a specific dilemma or problem”. As such, they continue, “*ambiguity, uncertainty, and disagreement are inevitable*” (p. 6). As suggested earlier, drawing upon pluralistic approaches foregrounds the notion that “multiple, ethically legitimate *judgement calls*” exist (p. 6). What is then needed is robust, reliable methods for determining and evaluating those calls, rather than relying on more rule-bound, “one size fits all” ethical and legal requirements. In what follows, we describe a range of approaches which foster ethical thinking throughout the design and development process, before outlining a range of tools and methods which can be used iteratively throughout the project’s life-cycle.

2.1 Value-Sensitive Design (VSD)

The VSD methodology involves conducting *conceptual*, *empirical*, and *technical* investigations. During the *conceptual* phase, stakeholder mapping exercises enable the identification of both direct stakeholders (those who interact directly with the technology) and indirect stakeholders (those who have only a peripheral relation with the technology). Within this phase, the general aim is to identify the values at stake in relation to the technology at hand, based on a broad analysis of vested interests.



Figure 2: Tripartite Methodology of VSD
(from Longo, Padovano, & Umbrello, 2020)

In the second *empirical* phase, both qualitative and quantitative analyses help shed light on how people interact with the technology and how different design choices might shape the prioritisation of competing values, as well as any potential value trade-offs. With regards to the *technical* phase, this involves analysing any technical issues with the design and considering, once again, the extent to which different design choices support or constrain the realisation of values identified during the conceptual phase.

The iterative phases of VSD recognise that aligning technology with the values, needs, and expectations of society is always an evolving process and that both technology and society play a role in shaping any possible outcomes (van de Poel, 2020). In other words, theoretically, VSD starts from the idea of the co-evolution of technology and society (see e.g. Rip & Kemp, 1998).

2.2 Virtual Ethnography & Ethnography of Design

Ethnography is the study of social interactions, behaviours, and perceptions that occur within groups, teams, organisations, and communities (Reeves & Hodges, 2008). According to Christine Hine, virtual ethnographies entail “developing appropriate levels of technical skill in order to participate effectively in the setting, but can also benefit from a step back from total technical competence in order to see the steps which precede effective entry and to bring into focus the taken-for-granted qualities of the setting” (2008, p. 263). Until relatively recently, the inaccessibility of VR has prevented the execution of immersive virtual ethnographies *in the wild*. In their study on harassment in social VR, for example, Ketaki Shriam & Raz Schwartz argue that more ethnographic research is required in order to study how technological features impact user behaviour within immersive environments.

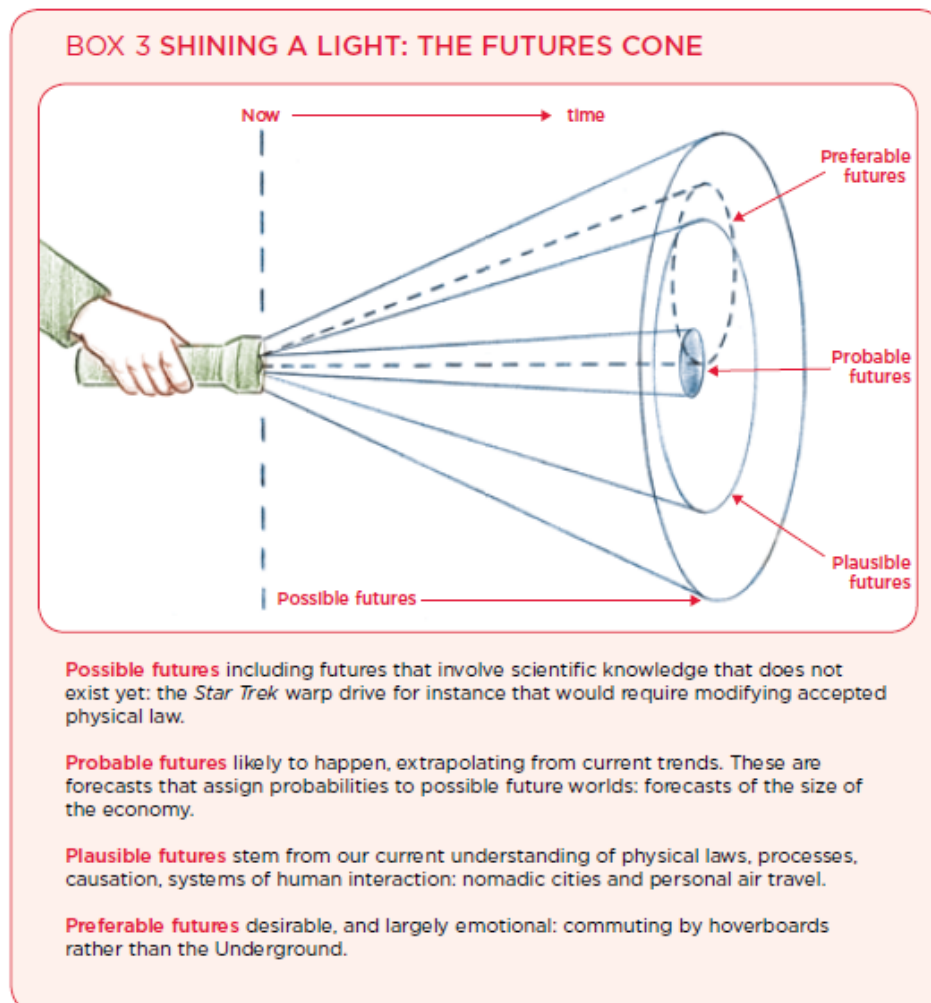
Whereas carrying out detailed ethnographic research within VR spaces is a relatively new possibility, there is already an established tradition of conducting ethnographic research within design teams interested in human-computer interaction (HCI).

However, according to Paul Dourish (2006), these studies have historically focused narrowly on deriving “implications for design”, rather than on encouraging genuine engagement between ethnographers and designers. Within intelligent virtual environments (IVEs), embedding ethnographers in design teams could enable them to trace and untangle the different practices and activities that encourage different values to manifest. For example, according to Katie Shilton (2012), various internal and external mechanisms, or “value levers”, can serve to open up discussion about the role of values in design. In her case, these value levers included “working on interdisciplinary teams, experiencing internal prototype testing, designing around constraints, advocacy by leaders and a values worker, navigating institutional mandates, and gaining funding” (p. 393), all of which played a different role in promoting social values throughout the design process. Her analysis of these value levers suggests that when leveraged appropriately they can result in, “(1) changes in the design conversation; (2) consensus around a social values as relevant and even useful to design; and (3) values-based modifications to the technologies themselves” (p. 391).



2.3 Futures Studies

Popular across the public, private, and social sectors, Futures Studies utilize both qualitative and quantitative methods in order to explore and propose possible, plausible, probable, and preferable futures—a taxonomy around since the late 1970s (see Figure 1).



*Figure 3: The Futures Cone
(from Bland & Westlake, 2013)*

Using a wide range of established and well-developed methods, futures studies are used to spot signals of change, track trends, and make projections. Ideas about the future are discussed and brought to life through scenarios, storytelling, and speculative design. Given the variety of tools available, the UK Government Office for Science's Future (2017) put together a toolkit which organizes these tools into practical purposes which they can serve:



1. Gathering intelligence about the future
2. Exploring the dynamics of change
3. Describing what the future might be like
4. Developing/testing policy and strategy

According to the UK's Innovation Foundation, NESTA, thinking about the future in a structured way is not just useful, but essential. They provide three maxims, each of which covers a different variant of Future Studies: “forecasting or calculating about futures”; “creating plausible future scenarios for foresight practices”; and “employing fiction in order to explore preferable (and un-preferable) futures” (p. 20).

NESTA's 3 Maxims

New forms of data-driven forecasting tell us valuable things about the near future. There is scope for experimenting with these techniques in order to find out what works.

Thinking about plausible future scenarios can help guard against fragility. Governments and businesses need foresight capabilities in order to address systemic challenges.

Innovation starts with a story about the future. Imagining and sharing desires and fears about the future is a way for all of us to shape it.

Speculative design fuses critical design and Futures Studies, focusing on the creation of ideas, prototypes, and experiments in order to imagine alternative products, systems, or worlds. Speculative design often involves mapping how the existing environment can inform future developments of technology, as well as engaging with counterfactual or alternate histories in order to question how and why things came to be the way they are and how they might have looked different. According to James Auger (2013), speculative design requires a “perceptual bridge” that allows people to shift their thinking from their ideas about the world as it is to fictional ideas about the world as it could be. He suggests that “inspiration and influence for this ‘perceptual bridge’ can be drawn from diverse fields such as observational comedy, psychology, ecology, horror films and illusion” (p. 12).

In the context of GuestXR, tools and methods drawn from VSD, virtual ethnography, Future Studies, and Speculative Design could help us to anticipate ethical and epistemic issues at an early stage in its development and therefore better align the development of IVRS with the needs, values, and expectations of society.

In the sections that follow we describe a number of tools which are drawn from the approaches outlined above. We have loosely organized these tools according to the iterative phases of Service Design: prepare, explore, understand, improve, and implement. It is important to stress that given the iterative nature of R&D however, these tools could potentially be used at any stage of the R&D process, or indeed repeated at various stages, depending on the topic, product, team, context, etc.

3 Ethics by Design Toolkit

3.1 Prepare

The preparation phase typically involves research design and problem formulation. During the preparation phase, key ideas are defined and theoretical and methodological frameworks are worked out. This phase also involves identifying relevant ethical frameworks, questions, and issues. The activities carried out in this phase help generate insights that will allow for operationalization of value consideration at later stages of the design and development process.

In the context of AI research, an important first step is to create an understanding of the AI, its operation, and social impact based on relevant contextual factors. Researchers must scrutinize the social, political, and economic context within which a technology operates—as well as the technology itself. As conditions and values already begin to surface during this phase, it is crucial to involve diverse stakeholders early on in the process, so that a wide perspective can be adopted, key values can be identified, and so as to avoid potentially problematic assumptions becoming embedded in the research process. Examples of key questions that should be reflected upon during and after the activities carried out in this phase are included at the end of this section.

3.1.1 Literature Review

A crucial part of the preparation phase is identifying relevant ethical frameworks, questions, and issues. Semi-systematic reviews map dominant themes that have emerged over time, drawing attention to how topics have developed across different research traditions. This allows for the synthesis of multiple perspectives across several areas of research. Using this approach provides a broad overview of the main issues, and helps to make sense of complex and often contested topics. A semi-structured review can then be used as a decision aid (see D5.1).

Given that the GuestXR project brings together research on VR and VR systems with AI and reinforcement machine learning techniques, as well as neuroscience and social psychology, deliverable D5.1 includes discussion of the following areas:

- the ethics of AI;
- ethical considerations related to the development and use of VR systems and IVEs, bringing ethical investigations of VR into dialogue with the ethics of social AI systems;
- ethics by design methodologies

The ethics state-of-the-art and literature review conducted for D5.1 served as the knowledge base for development of these guidelines.

3.1.2 Stakeholder Analysis

In order to understand the values at stake within any project, engaging the relevant stakeholders is an important first step. Used as a business strategy, stakeholder analysis enables a greater understanding of a product or service's audience, allowing businesses to tailor their business plan and product launch accordingly.

Within the context of VSD and ethics by design approaches, stakeholder analysis involves the identification of individuals, groups, organizations, institutions, and societies who might impact, or be impacted by, the technology (both positively and negatively).

Two overarching stakeholder categories include: 1) direct stakeholders (in this case, those who will interact directly with the technology), and 2) indirect stakeholders (those only indirectly affected by the technology) (Kahn, Hagman, Severson, & Gill, 2006; Nathan, Friedman, Klasjna, Kane, & Miller, 2008; Czeskis, Dermendjieva, Yapit, Borning, Friedman, Gill, & Kohno, 2010).

Step 1: Identify stakeholders

The first step in a stakeholder analysis is to draw up a list of relevant actors. These might include funders/investors, government agencies, universities, media, consumers, regulatory authorities, etc. As already mentioned, essentially anyone who might impact, or be impacted by the technology.

Useful questions to ask when drawing up such a list might include: who will be affected? Who will support the project? Who will oppose it? Other useful strategies for identifying relevant actors might include brainstorming, focus groups, consultations, research, and historical data.

Step 2: Characterization of Stakeholders

The next step is to loosely characterize actors according to their interests and needs. Putting together profiles of actors or actor personas can be a helpful tool in characterizing actors (you can find downloadable templates for carrying out such as exercise here: <https://servicedesigntools.org/tools/personasvarious> and here <https://medium.muz.li/what-are-how-to-create-personas-step-by-step-guidelines-of-everything-49357da2cb59>). In addition, visualisation tools can also be used to enable grouping actors together in stakeholder groups e.g. Venn diagrams and Net-Maps (See Lelea et al, 2014).

Step 3: Analyze Stakeholders

Once identified and categorized, the stakeholders' power, resources, values, and relationships can be assessed in more detail. Table 3 provides examples of the types of issues, goals, and questions considered part of a stakeholder. This step analysis a better understanding of the people involved, the issues that matter to them, and their expectations and needs. These questions are typically addressed during dialogues and interviews with stakeholders (see 3.1.3 Interviews & Surveys).



Issue	Goal(s)	Questions to be addressed
Power	<i>Examine power differences between actors that may be important for the issue/problem to be addressed</i>	<i>How does the actor influence other actor's role and scope of action?</i> <i>How is the actor influenced by other actors with regard to his/her role and scope of action?</i> <i>What is the actor's authority in terms of:</i> <i>Setting objectives and norms?</i> <i>Allocating or denying resources to other actors?</i> <i>Defining others' tasks and responsibilities?</i> <i>Controlling access to knowledge/information?</i> <i>Allocating rewards/recognition/sanctions?</i> <i>Channelling messages to superiors and external bodies?</i>
Interest	<i>Describe the actors' interest(s) in relation to the problem/issue the project focuses on</i>	<i>What is the actor's interest in the issue/problem?</i> <i>What prior experience does the actor have on the problem/issue to be addressed?</i> <i>What potential benefits and risks arise for him/her if the issue is addressed/the problem is solved?</i> <i>To which degree are the actor's interests coherent or conflictive with other actors' interests?</i> <i>What options exist to increase the actor's interest and engagement, or to dismantle obstacles?</i>
Resources	<i>Identify material and non-material resources different actors possess or have control over</i>	<i>• What relevant resources does the actor have at his/her disposal (e.g. knowledge, expertise, skills, and material resources)?</i> <i>What potential contributions could arise from this?</i> <i>Which resources are missing that may be needed to effectively address the issue/solve the problem?</i>
Relationships	<i>Describe the relationships between actors, e.g. with regard to distance, degree of trust and/or conflict</i>	<i>Which actors cooperate with each other?</i> <i>Which actors compete with each other?</i> <i>Which actors tend to be disconnected from other actors?</i> <i>Which degree of trust/mistrust is there between the various actors?</i> <i>Which conflicts exist and how explicit are they?</i>
Social difference	<i>Identify differences in actors' power, interest, resources and relationships that relate to social categories they belong to</i>	<i>How are differences in power, interest, resources and relationships related to age, gender, ethnic group, educational background or wealth?</i> <i>Are any strategies necessary to avoid exclusion of certain groups of actors?</i> <i>Which empowering measures may be required?</i>

Table 3: Issues, Goals, and Questions to be addressed during Stakeholder Analysis
(from Lelea et al., 2014)



During the stakeholder analysis, it is also possible to further classify stakeholders according to various typologies. The most common differentiation is between “key”, “primary”, and “secondary” stakeholders (Table 4). Some stakeholders might fit across multiple categories, or change categories through the course of the project. This is why stakeholder analysis should be considered iterative and repeated at various stages throughout the project’s life-cycle (Lelea, 2014).

Stakeholders	Description
Primary Stakeholders	<i>Primary stakeholders are the people most affected.</i>
Secondary Stakeholders	<i>Secondary stakeholders can be harder to identify because they are indirectly affected and their motivations may therefore be less obvious.</i>
Key Stakeholders	<i>Key stakeholders are the people who have the most power or influence; they are generally considered vital to success. Not only do they have a vested interest in the outcome, but they have the power to seriously impact its delivery as well.</i>

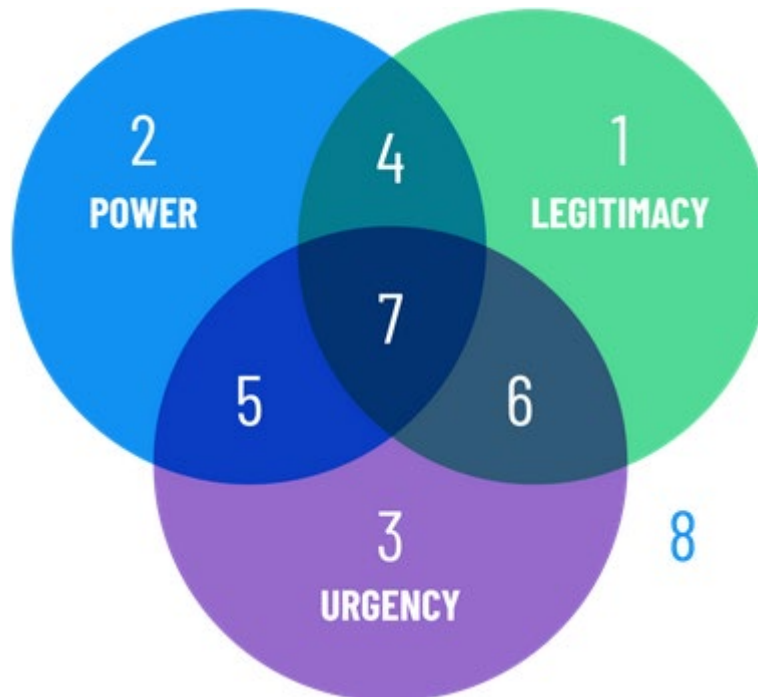
Table 4: Description of Stakeholder Types

Step 4: Map stakeholders

Once the various stakeholders’ issues, goals, values and needs have been analysed they can then be plotted on a grid. Their position on the map helps to illustrate what is at stake and highlight the need for engagement strategies throughout the project. While from a business point of view, the power/interest graph is commonly utilised, from an ethics by design perspective the salience model (figure 3) provides greater nuance, assigning stakeholders to one of seven classes, which are prioritized based on their levels of **power, legitimacy, and urgency**. All stakeholders fall into three overlapping circles, as in the accompanying Venn diagram. Those who fit into more than one category (indicated by the overlapping sections) require the most attention (Patrick, 2021).

As pointed out by Bechmann et al. (2020), issues surrounding power and legitimacy are “particularly pertinent when dealing with AI as the learning models may show patterns that are unexpected” and because “AI can also amplify already existing social hierarchies”. As they make clear, “the influence of stakeholders (e.g. developers in different communities and organizations inside and outside academia) may be strong when for instance cleaning data, using pre-trained models or changing the models” (p. 36).





*Figure 3: The Salient Model
(from Patrick, 2021)*

Stakeholder mapping is an ongoing exercise. Though the first analysis should occur in the preparation phase, it is important to re-evaluate stakeholder mapping regularly throughout the course of any project. As projects unfold, stakeholders come and go, meaning that new value conflicts may emerge as things change.

Useful activity outlines, templates and instructions for stakeholder analysis are available via the Co-Creation Navigator: <https://ccn.waag.org/navigator/theme/stakeholders>

3.1.3 Stakeholder Tokens

Stakeholder Tokens can be used as an additional tool to help in both the identification of, and engagement with, stakeholders. The goal of Stakeholder Tokens is threefold (from Yoo, 2018):

- 1) To generate a more inclusive set of stakeholders, i.e. by surfacing hidden, overlooked, or neglected stakeholders;
- 2) To identify a robust set of key stakeholders, i.e. by providing an empirical rationale for their inclusion;
- 3) To clarify stakeholder dynamics, i.e. within a complex (and often conflictual) socio-political setting.

According to Yoo, as a method Stakeholder Tokens are **participatory, visual & tactile, and creative & Playful**. Inspired by LEGO serious play (Cantoni et al. 2009), Stakeholder Tokens involve:



Step 1: Select Participants

Invite participants based on a list of both direct and indirect stakeholders.

Step 2: Select Token Materials

A token represents a stakeholder. Tokens should be small enough to hold and move around on a table-top. Tokens should also have a unique identifier (such colour, shape, or texture). Yoo recommends preparing around 10 to 20 tokens per session.

Step 3: Create a List and Labels

Participants are asked to identify relevant stakeholders and create lists and labels.

Participants should be prompted to consider stakeholders who may typically be left out.

Useful prompts
Who are the important people, groups, or communities involved? Who else do you think would care about this issue and why? Is there anyone who is left out?

Step 4: Attach labels to tokens

Participants use the tokens to represent different stakeholders.

Step 5: Sketch stakeholder interrelationships

Participants are asked to place tokens on a large piece of paper and to use pens to sketch the relationships between them.

Participants should be prompted to consider what information, resources etc. travel between stakeholders and to enact short scenarios in order to further illustrate the interactions between them.

Useful prompts
What are the relationships among these stakeholders? Please make a drawing to show their relationships. Please act out a short scenario using your tokens to illustrate some of these relationships.

Stakeholder Tokens can be utilized to engage stakeholders in the process of stakeholder analysis itself, providing a useful tool for conducting empirical investigations.



3.1.4 Value Source Analysis

During the preparation phase, it is also important to distinguish between the explicitly supported project values, designers' personal values, and values held by other direct and indirect stakeholders (Friedman, Hendry & Borning, 2017). Crucially, explicitly supported values and designers' personal values are not the same. The former are "subjected to a principled analysis of arguments for their inclusion rather than simply being a matter of personal preference" (Borning et al., 2005). During value source analysis, dialogues and semi-structured interviews again play an important role, as do reflexive team evaluations and questionnaires. As Borning et al. (2005) point out, it is important that no technology favor any given set of values *a priori*. Instead, multiple stakeholders should be encouraged to articulate the values that are important to them in order that a full evaluation of relevant values can be made.

3.2 Explore

The exploration phase typically entails activities related to data collection and experimental testing. During the exploration phase, values and needs should be examined in terms of how they emerge, change, or evolve, as well as what sorts of tensions or trade-offs appear at different stages and why. During the exploration phase it is important to examine the ideas and experiences of a variety of stakeholders, both within and beyond the project.

Applying the IRE 3.0 to AI and machine learning (ML), Bechmann et al. (2020) suggest that during the exploration phase, researchers who use AI systems in their research consider carrying out their typical *ex ante* hypothesis (while still making room for exploration), supplementing this by documenting the actions taken and choices made throughout the process along with self-reflections on how this research practice has affected their findings and research questions (p. 38). Examples of key questions that should be reflected upon during and after the activities carried out in this phase are included at the end of this section.

3.2.1 Interviews & Surveys

Stakeholder interviews provide data about people's beliefs, perspectives, and meaning-making (Roulston & Choi, 2018). Semi-structured interviews are a way of asking questions that delve into stakeholders' understandings, views and values with regards to a specific technology. Questions can be asked so as to elicit stakeholders' evaluative judgments about a technology as well as their rationale as to how they arrived at that evaluation.

Semi-structured interviews typically follow a topic guide which is a logical sequence of topics and questions that help the researcher conduct an interview session. The guide is organized before the interview according to the different topics that need to be explored during the research; each section comes with a detailed set of questions that help cover that topic. During interviews, conversation may often go in many different directions, a clear topic guide helps researchers remember all the key aspects to explore. (See <https://servicedesigntools.org/tools/interview-guide> for tips on creating a topic guide)

Semi-structured interviews can be preceded or supplemented by conducting surveys of a broader selection of stakeholders.

Interviews and surveys can be utilised at all stage of the R&D process, from preparation right through to implementation. Within GuestXR, initial interviews were carried out with WP leaders during the preparation of D.5.1.

D.5.2 provides a script for stakeholder interviews and a survey that can be used in conjunction with these guidelines and throughout the remainder of the project.

3.2.2 Focus Groups

In addition to conducting individual interviews, another useful strategy can be hosting a focus group. During a focus group, researchers invite a group of people and ask them questions about specific products, services, goods, concepts, problems, prototypes, advertisements, etc.

The goal of a focus group is to get a broader understanding of the perceptions, opinions, ideas, or attitudes of the group toward a given topic. Focus groups are mostly carried out in informal settings. The idea is that participants discuss the given topics from their own perspective. Researchers therefore typically start the discussion, before taking a step back and observing the group discussion and dynamics. Sometimes a researcher acts as a moderator, guiding the group through a set of questions. In a dual-moderator focus group one researcher facilitates the process while the other observes. In contrast to co-creation workshops, researchers do not act as facilitators and the participants do not work with boundary objects in order to create an outcome together.

Step 1: Define Specific Research Question

Step 2: Recruit Participants

Step 3: Plan and Prepare

Step 4: Conduct Focus Groups

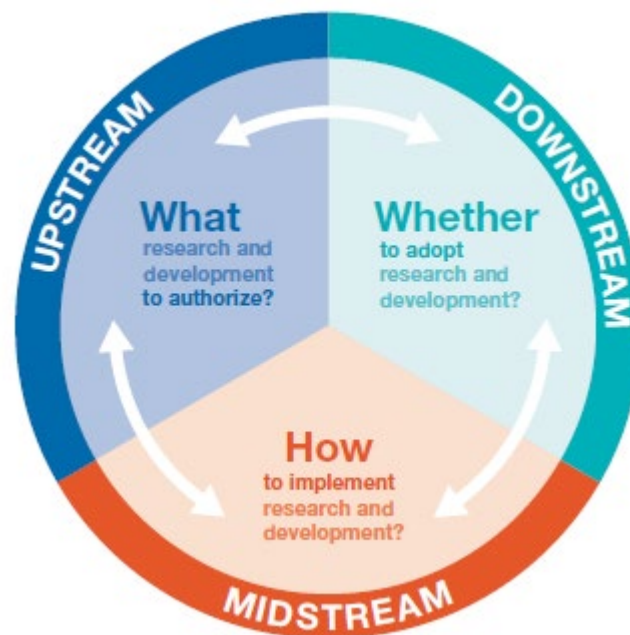
Step 5: Follow-up

See <https://www.thisisservice.designdoing.com/methods/focus-groups> for a detailed explanation of these steps.

3.2.3 Midstream Modulation

The rise of the concept of Responsible Innovation reflects the growing need for ethical considerations to be integrated within R&D processes. Whereas in the past ethics was often either an *upstream* policy question (i.e. asking whether or not certain research should be carried out) or part of a downstream evaluation/assessment (in terms of whether and how the R&D would be adopted), the approach of Midstream Modulation (MM) stresses the importance of introducing different “modulations” (changes to the research process and its results), during the “midstream” phase, which denotes “the phase of research and development before scientific results are translated into products or services, but after authorization and funding decisions have been taken” (Schuubiers & Fisher, 2009, p. 425) (see figure 4)

MM utilizes ethnographic methods for data collection and analysis to uncover the complex relationships among values, technology and social structure as those relationships unfold during research.



*Figure 4: Stages in the Governance of Science & Technology
(from Schuurbiers & Fisher, 2009, p. 425)*

As Schuurbiers and Fisher (2009) suggest, rather than seeing the inclusion of ethical thinking and the need to consider societal considerations as “ethical speed bumps”, MM is designed to “evaluate and adjust research decisions in light of social factors while the research process is taking place” (Schuurbiers & Fisher, 2009, p. 425). As a form of Socio-Technical Integration Research (STIR), MM typically involves in-depth engagement in situated contexts over longer periods of time. STIR requires embedding social scientists (sociologists, political scientist, etc.) or humanists (philosopher, ethicists, etc.) in teams with technical practitioners, providing a framework through which they carry out intervention-oriented activities aimed at elucidating and enhancing the responsive capacity of practitioners to the broader societal dimensions of their work (Fisher et al., 2006).

Through collaborative exploration of research decisions, practices, organization, etc. the main aim of STIR is to shape technological trajectories by rethinking the processes that are typically taken for granted (Schuurbiers & Fisher, 2009).

One approach to MM is the use of a “decision protocol” (figure 5). During their time embedded in the team, the STIR scholar closely follows and documents the research process. Using the decision protocol, the STIR scholar meets regularly with different members of the team to discuss and explore decision moments, and to encourage reflection on perceived alternative and potential outcomes.



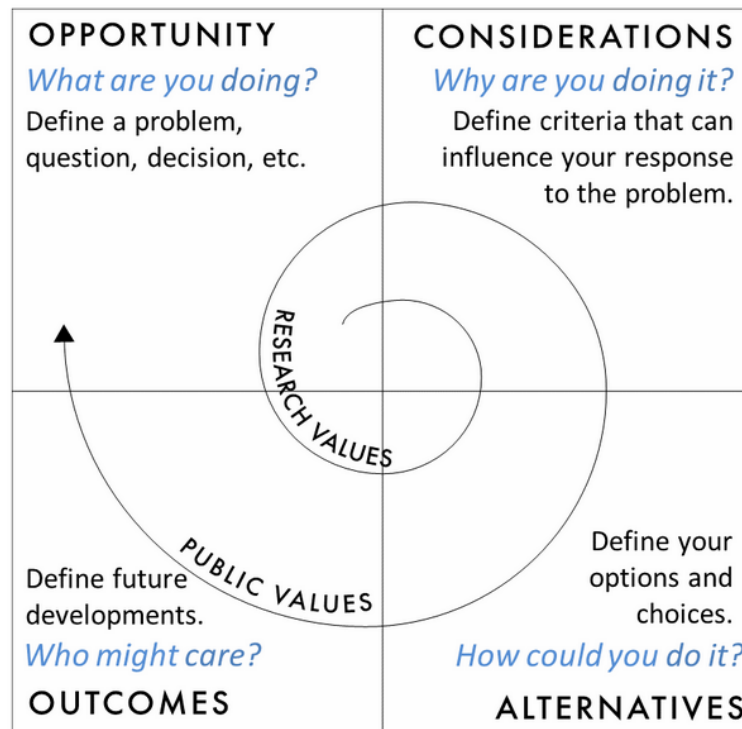


Figure 5: The Decision Protocol
 (from <https://cta-toolbox.nl/tools/STIR/>)

The MM protocol will be used to guide the research stay of the UM ethics team in M13 + M14 of GuestXR.

3.2.4 Value Scenarios

Value scenarios support the envisioning of the systemic effects of new technologies (Nathan et al., 2005). Value scenarios are narratives, comprised of stories of use, which emphasize the value considerations of new technologies, particularly with regards to the long-term and potentially negative impacts. Value scenarios can be written by any stakeholder as long as they emphasize the implications for both direct and indirect stakeholders equally. Value scenarios are not an attempt to predict the future, but provide a means for thinking about the ways in which decisions and actions made in the present will shape the conditions of the future.

According to Nathan et al. (2005), value scenarios draw upon 5 key elements in order to develop provocative sketchers of the future: stakeholders, pervasiveness, time, systemic effect, and value implications (Table 5).



Stakeholders	Description
Stakeholders	<i>Following Value Sensitive Design, value scenarios help designers envision a range of effects of a pervasive technology, both on those who are in direct contact with a technology (direct stakeholders), and on those who might not be direct users, but whose lives are affected by various interactions around the technology (indirect stakeholders)</i>
Pervasiveness	<i>A value scenario presents a vision in which a technology has become widespread, spanning various geographic regions, cultures, social classes, and other contexts (e.g. school, work, home, car).</i>
Time	<i>Rather than focus on short-term effects, value scenarios take into consideration what the world might look like five, ten, or twenty years after a technology has been deployed.</i>
Systemic Effects	<i>Value scenarios explore the multidimensional interactions among technology, psychology, society, culture, and the environment as use of the technology becomes pervasive over a period of years.</i>
Value Implications	<i>Finally, drawing on VSD and aspects of design noir (ref), value scenarios help envision not only positive effects of technology, but also its darker consequences. We suggest that a careful consideration of a diverse range of influences, including the negative, should be a key component of the design process.</i>

*Table 5: 5 Key Elements of Scenario Building
(from Nathan et al., 2005)*

As Nathan et al (2005) argue, “Through value scenarios, designers and policy makers can begin to imagine a wide range of influences for a proposed technology. The design team and/or policy makers are assisted in considering various appropriations which can be supported or constrained by system and policy implementations”.

A few rich, nuanced scenarios can help to focus attention on both direct and indirect stakeholders, value tensions, and long-term societal implications that might otherwise go unnoticed (Czeskis et al., 2010). Like most of the tools described in this toolkit, value scenarios can be utilized at various stages throughout the R&D process.

The UM team will run a scenario workshop as part of a GuestXR consortium meeting in 2023. When the workshop protocol is prepared it will be included in D5.2.



3.2.5 Consequence Scanning

Consequence Scanning is similar in aim to value scenarios in that it allows you to consider the potential consequences of what you are building and provides opportunities to mitigate or address potential harms before they happen (Brown, 2019). Consequence Scanning is based upon the discussion of three questions:

1. What are the intended and unintended consequences of this product or feature?
2. What are the positive consequences we want to focus on?
3. What are the consequences we want to mitigate?

These questions are designed in order to help teams share knowledge and expertise and map the potential impacts of what they are working on. According to Brown (2019), Consequence Scanning can “spark new ideas and foster positive conversations and common understanding between all stakeholders.

As with many of the tools presented in this toolkit, Brown highlights that Consequence Scanning is most effective when participants are cross-functional and multi-disciplinary. Including a broad array perspectives and expertise provides a greater opportunity to identify potential consequences.

Intended consequences	Unintended consequences
<i>Refers to the change or impact you are looking to make.</i>	<i>Refers to what could happen as a result of your actions.</i>

Brown provides context prompts to encourage teams to think about potential consequences beyond direct user interactions, including potential impacts on individuals and communities, as well as consequence prompts, categories and case studies of unintended consequences which commonly feature within discussions regarding digital technologies (download Consequence Scanning prompt cards and manual here: <https://doteveryone.org.uk/project/consequence-scanning/>)

3.2.6 Backcasting

Backcasting is another tool for reflecting on how our choices in the present potentially affect the future. Usually conducted as a group activity during a workshop, Backcasting involves coming up with one or more hypothetical future scenarios and working backwards to identify the steps needed to connect different possible futures to the present. In this way participants can reflect on different directions in which specific organisations, products, services or initiatives may evolve, and extract considerations able to affect their work today—so as to articulate what a desirable and undesirable future might look like.



Step by Step Guide to Backcasting (from <https://servicedesigntools.org/tools/future-backcasting>)

Step 1: Select a Topic

Select a topic or specific service industry, get inspired by recent news (signals) about it and decide how many and what types of future scenarios you would like to shape (Possible, Plausible, Probable or Preferred). Once you have defined one or more future scenarios, set a hypothetical year and briefly describe each of them in the form of an article (title + short description).

Remember to make available a couple of links/news from the present to make people understand where the future scenarios come from.

Step 2: Build the Narrative

Build the narrative to describe the “today section” selecting the best tool between personas, service description and system scenario.

Remember to highlight all the useful information that could affect the future scenario.

Step 3: Define Steps

Define one or more steps needed to connect each future scenario to the present. For each step, specify the blockers and enablers in order to easily identify the most critical points.

Step 4: Identify Pathway

Identify the quick wins that help you to get closer to the future scenario.

3.2.7 Risk Mitigation

The Ethical Operating System developed by the Institute for the Future and the Omidyar Network (2018) includes a risk mitigation tool designed to facilitate better product development. The Risk Mitigation tool presents eight risk zones where hard-to-anticipate and unwelcome consequences are likely to emerge.



Figure X: 8 Risk Zones
(from *Institute for the Future, 2018*)

Step 1: Choose

Choose a technology, product, or feature you are working on

Step 2: Imagine

Imagine scenarios for each risk zone (see examples of scenarios for each case in the Ethical Operating System Manual)

Step 3: Identify

Identify the checklist questions most relevant to the technology you've selection (see table X)

Step 4: Think

Think about how you might start to correct or mitigate the risk you've identified

Download the full manual here: <https://ethicalos.org/>



Risk Zone	Questions to be addressed
Truth, Disinformation, and Propaganda	<i>What type of data do users expect you to accurately share, measure or collect?</i> <i>How could bad actors use your tech to subvert or attack the truth?</i> <i>What could potentially become the equivalent of fake news, bots or deepfake videos on your platform?</i> <i>How could someone use this technology to undermine trust in established social institutions, like media, medicine, democracy, science?</i> <i>Could your tech be used to generate or spread misinformation to create political distrust or social unrest?</i> <i>Imagine the form such misinformation might take on your platform. Even if your tech is meant to be apolitical in nature, how could it be co-opted to destabilize a government?</i>
Addiction & the Dopamine Economy	<i>Does the business model behind your chosen technology benefit from maximizing user attention and engagement—i.e., the more, the better? If so, is that good for the mental, physical or social health of the people who use it? What might not be good about it? What does “extreme” use, addiction or unhealthy engagement with your tech look like?</i> <i>What does “moderate” use or healthy engagement look like?</i> <i>How could you design a system that encourages moderate use?</i>
Economic & Asset Inequalities	<i>Who will have access to this technology and who won’t?</i> <i>Will people or communities who don’t have access to this technology suffer a setback compared to those who do?</i> <i>What does that setback look like?</i> <i>What new differences will there be between the “haves” and “have-nots” of this technology?</i> <i>What asset does your technology create, collect, or disseminate? (example: health data, gigs, a virtual currency, deep AI) Who has access to this asset?</i> <i>Who has the ability to monetize it? Is the asset (or profits from it) fairly shared or distributed with other parties who help create or collect it?</i> <i>Are you using machine learning and robots to create wealth, rather than human labor?</i>
Machine Ethics & Algorithmic Biases	<i>Does this technology make use of deep data sets and machine learning?</i> <i>If so, are there gaps or historical biases in the data that might bias the technology?</i> <i>Have you seen instances of personal or individual bias enter into your product’s algorithms?</i> <i>How could these have been prevented or mitigated?</i> <i>Is the technology reinforcing or amplifying existing bias?</i> <i>Who is responsible for developing the algorithm?</i> <i>Is there a lack of diversity in the people responsible for the design of the technology?</i>

	<i>How will you push back against a blind preference for automation (the assumption that AI-based systems and decisions are correct, and don't need to be verified or audited)?</i> <i>Are your algorithms transparent to the people impacted by them?</i> <i>Is there any recourse for people who feel they have been incorrectly or unfairly assessed?</i>
Surveillance State	<i>How might a government or military body utilize this technology to increase its capacity to surveil or otherwise infringe upon the rights of its citizens?</i> <i>What could governments do with the data you're collecting about users if they were granted access to it, or if they legally required or subpoenaed access to it?</i> <i>Who, besides government or military, might use the tools and data you're creating to increase surveillance of targeted individuals?</i> <i>Whom would they track, why—and do you want your tech to be used in this way?</i> <i>Are you creating data that could follow users throughout their lifetimes, affect their reputations, and impact their future opportunities?</i> <i>Will the data your tech is generating have long-term consequences for the freedoms and reputation of individuals?</i> <i>Whom would you not want to use your data to surveil and make decisions about individuals, and why not?</i> <i>What can you do to proactively protect this data from being accessible to them?</i>
Data Control & Monetization	<i>Do your users have the right and ability to access the data you have collected about them?</i> <i>How can you support users in easily and transparently knowing about themselves what you know about them?</i> <i>If you profit from the use or sale of user data, do your users share in that profit?</i> <i>What options would you consider for giving users the right to share profits on their own data?</i> <i>Could you build ways to give users the right to share and monetize their own data independently?</i> <i>What could bad actors do with this data if they had access to it?</i> <i>What is the worst thing someone could do with this data if it were stolen or leaked?</i>

*Table 6: Risk Mitigation Checklist
(from Institute for the Future, 2018)*



3.3 Understand

Once data has been collected, analysis is conducted in order to generate insights which can be operationalized. In the context of AI research, analytical processes include selecting the training data, cleaning the data, developing the model through steps of training, evaluating, adjusting, and re-training the model (Bechmann et al., 2020). It is therefore during this phase that values need to be translated into design requirements.

It is worth pointing out that within AI research, analyses during this phase also include assessments of how particular techniques, formulas, or instruments may re-identify data through aggregation of multiple data-sets. This means that downstream ethical effects arising from the unpredictability of algorithmically driven analytical processes also need to be considered (AoIR, 2019).

3.3.1 Value-Norm-Requirement Hierarchy

Most of the tools described so far support the conceptual and empirical investigation of VSD, in that they can be used in order to identify the values at stake. However, values are typically abstract philosophical concepts that are difficult to operationalize in terms of concrete engineering requirements. Within VSD, norms are considered to occupy the “transition point between values and design requirements” and therefore represent “the contextual designations of those values”, i.e. norms can be understood as the “design objectives of any given project (i.e. ‘maximize safety/usability/efficiency’ or ‘minimize cost’)”. The value-norm-requirement hierarchy (figure 6) illustrates the bi-directional process through which designers can begin to conceptualize which design requirements support what norms.

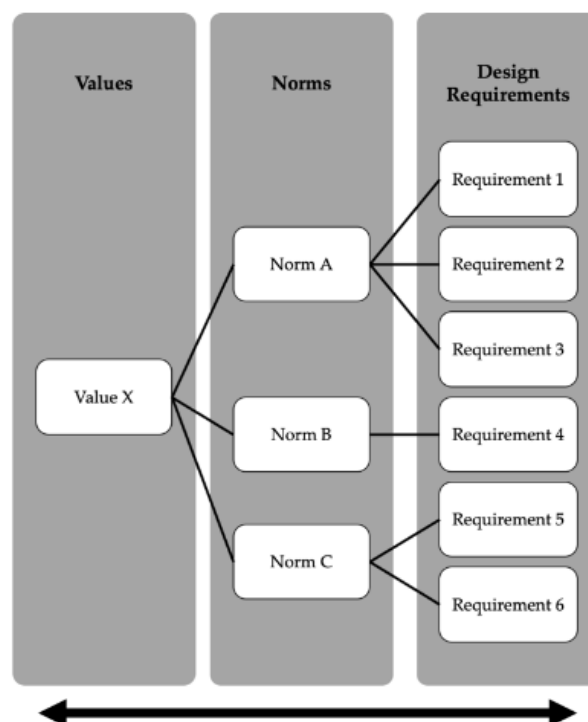


Figure 6: Bi-directional value-norm-requirement hierarchy
(from Longo, Padovano, & Umbrello, 2020)



Crucially, as Longo et al. point out, there is not one avenue from value to requirements, there are numerous ways in which values can become embodied in design. This is why the hierarchy needs to be explored from both directions, and technical teams need to specify how values, norms, and requirements relate to each other in within the given project.

The design requirements identified must be continuously checked for corresponding value and norm alignment and technical feasibility throughout the project. The reflection gates offer suitable moments to carry out such checks. Alternatively, these reflections can also take place implicitly by being included or “built in” within other co-creation activities.

3.3.2 Value Dams and Flows

Another tool for translating the results of stakeholder analysis into specific, implementable design features is the Value Dams and Flows method. Developed by Miller et al. (2007), The Value Dams and Flows method works by:

- Avoiding features that even a small number of stakeholders view as particularly problematic
- Identifying and designing for values stakeholders wish to see the system embody
- Systematically addressing values-oriented design trade-offs.

Dams and Flows refer to the technical features that are disliked by even a small majority of stakeholders (dams) and those which are liked by many stakeholders (flows). The technique enables the identification of viable solutions from among a range of possible designs, helping to resolve value tensions among design choices. Options that are identified as dams can then be removed from the design space, whereas options identified as flows can be shortlisted as good candidates for the design solution. Czeskis et al (2010), extend the values and dams technique to identify what data to collect (or not collect), and of the collected data, what to reveal to whom and under what conditions.

3.3.3 Evaluation Matrix

Another means for translating abstract values into clear design requirements is using an evaluation matrix. The evaluation matrix allows researchers to prioritize different ideas, rating them based on a set of defined criteria, in order to identify the most promising ones. A common set of criteria includes the level of complexity related to idea implementation, and level of value they will bring to the user and to the organization (see <https://servicedesigntools.org/tools/evaluation-matrix> and Stevanovic, Marjanovic & Storga, 2015).



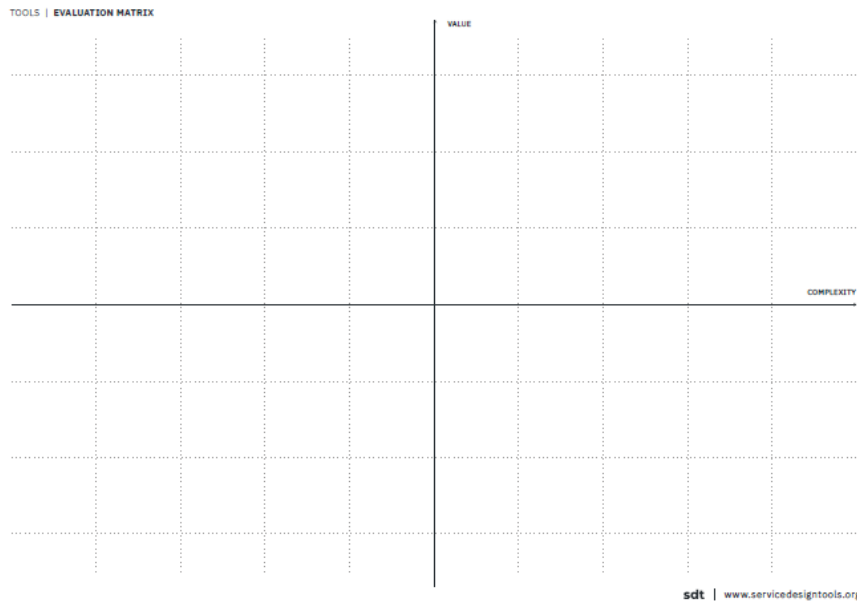


Figure 7: Evaluation Matrix
 (from <https://servicedesigntools.org/tools/evaluation-matrix>)

3.4 Improve & Implement

In this phase concepts, services, and prototypes are developed, tested, and implemented. Many of the tools already discussed can again be applied here, such as interviewing, surveys, focus groups etc.

3.4.1 User Stories

Like many of the tools outlined so far, user stories can be used at various stages throughout the R&D process. Building upon the insights generated during stakeholder analysis, they are particularly useful during prototyping and implementation. User stories allow the team to agree on which stories need to be part of the first prototype, to test selected stories, and to agree in which sequence the stories should be implemented (see 8). User stories are typically used in software development to define requirements from a user or customer perspective, in contrast to often rather product-based requirement documents.

OFTEN, **USER STORIES** ARE
FORMULATED LIKE THIS:

As a

..... (type of user/persona/role),

I want

..... (action),

so that

..... (outcome).

Figure 8: Template for Generating User Stories
(from <https://www.thisisservice.designdoing.com/methods/writing-user-stories>)

User stories should be formulated without IT-specific language. They should be written from the user's perspective, using simple, concise words. In service design, user stories are used to connect design research with actionable input for IT development. User stories can also be used during prototyping and particularly during implementation to turn low-fidelity prototypes into working software.

See *This is Service Design Doing* methods library:
<https://www.thisisservice.designdoing.com/methods>



3.4.2 Value-Oriented Mock-up, Prototype, or Field Deployment

This refers to the development, analysis, and co-design of mockups, prototypes and field deployments to scaffold the investigation of value implications of technologies that are yet to be built or widely adopted (Friedman et al., 2013). This involves identifying key technical challenges and creating architectural hooks in order to support desirable values and avoid some of the potential negative impacts of the system. The development of these strategies is dependent upon earlier empirical analysis (Czeskis et al., 2010). Mock-ups, prototypes or field deployments emphasize implications for direct and indirect stakeholders, value tensions, and technology situated in human contexts and can be introduced into development, analysis, and co-design processes (Friedman et al., 2017). This might include low-fi mock-ups, video prototypes, paper prototypes, “wizard of Oz” prototypes, working prototypes, and field deployment.

3.4.3 Value Sensitive Action-Reflection Model

In co-design and similar types of activities, a common challenge is to position stakeholders to generate creative ideas or to reflect on their ideas (Sanders and Westerlund, 2011). The Value Sensitive Action-Reflection Model blends VSD methods in order to address this challenge in a structured, reflective, and iterative manner (Friedman et al., 2017). Building upon traditional ideas about co-design—creative, flexible spaces that afford co-created responses to an initial design challenge—the model introduces stakeholder’s prompts and designer’s prompts as two types of value sensitive intervention. A variety of materials can function as prompts, for stakeholders these could be value scenarios, while for designers these could be user stories or personas. The interventions can be used multiple times and in any order, including the use of only stakeholders or only designer prompts. The novelty of the approach is that it supplements established practices of co-design, which typically focus on user’s needs and desires, with a focus on values and societal concerns (Yoo et al., 2013).

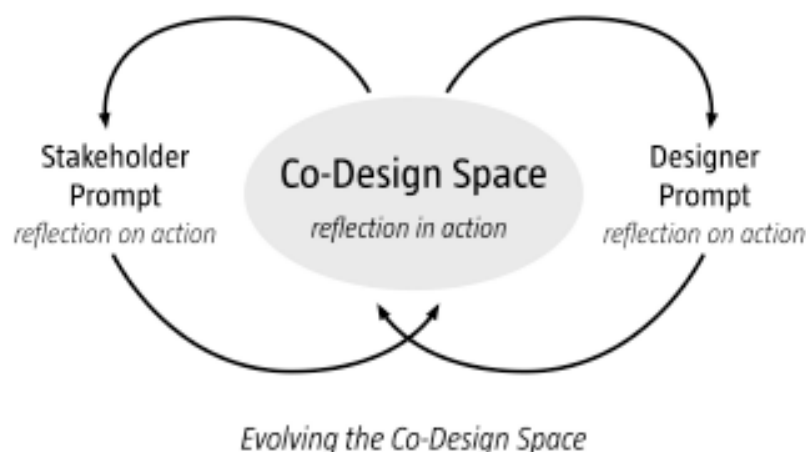


Figure 9: The Value-Sensitive Action-Reflection Model
(from Yoo et al., 2013)



4 CONCLUSIONS

This document contains a broad overview of ethics by design tools and methodologies. The idea of the document is to stimulate interest and reflection on these tools, as well as to provide information regarding the sorts of activities that partners may be encouraged to participate in at various stages throughout the project. The document can therefore be seen as a sort of toolkit, bringing together a range of different approaches in order to showcase what ethics by design is all about. Partners are encouraged to read the document thoroughly and approach the UM team with any questions or suggestions they may have. Partners are also encouraged to return to the document regularly, both because different tools may come into play at different stages in the project, as well as because the document will be continually updated based on the activities that are undertaken.

5 REFERENCES

- [1] AOIR (2019). Internet Research: Ethical Guidelines 3.0. Retrieved December 2, 2022, from <https://aoir.org/reports/ethics3.pdf>
- [2] Attard-Frost, B., De los Ríos, A., & Walters, D. R. (2022). The ethics of AI business practices: a review of 47 AI ethics guidelines. *AI and Ethics*, 1-18. <https://doi.org/10.1007/s43681-022-00156-6>
- [3] Auger, J. (2013). Speculative design: crafting the speculation. *Digital Creativity*, 24(1), 11-35. <https://doi.org/10.1080/14626268.2013.767276>
- [4] Bechmann, A. & Zevenbergen, B. (2020). AI and Machine Learning: Internet Research Ethics Guidelines, IRE 3.0 Companion 6.1, Association of Internet Researchers. Retrieved December 2, 2022, from <https://aoir.org/reports/ethics3.pdf>
- [5] Bland, J. & Westlake, S. (2013). Don't stop thinking about tomorrow: A modest defence of futurology. NESTA. Retrieved December 2, 2022, from <https://www.nesta.org.uk/report/dont-stop-thinking-about-tomorrow-a-modest-defence-of-futurology/>
- [6] Blok, V., & Lemmens, P. (2015). The emerging concept of responsible innovation. Three reasons why it is questionable and calls for a radical transformation of the concept of innovation. In *Responsible innovation 2* (pp. 19-35). Springer, Cham.
- [7] Borning, A., Friedman, B., Davis, J., & Lin, P. (2005). Informing public deliberation: value sensitive design of indicators for a large-scale urban simulation. In *ECSCW 2005* (pp. 449-468). Springer, Dordrecht.
- [8] Brown S. (2019). Consequence Scanning Manual: Version 1. London: Doteveryone. Retrieved December 2, 2022, from https://doteveryone.org.uk/wp-content/uploads/2019/04/Consequence-Scanning_Agile-Event-Manual_TechTransformed_Doteveryone.pdf
- [9] Burget, M., Bardone, E., & Pedaste, M. (2017). Definitions and conceptual dimensions of responsible research and innovation: A literature review. *Science and engineering ethics*, 23(1), 1-19. <https://doi.org/10.1007/s11948-016-9782-1>
- [10] Cantoni, L., Botturi, L., Faré, M., & Bolchini, D. (2009). Playful holistic support to HCI requirements using LEGO bricks. In *International Conference on Human Centered Design* (pp. 844-853). Springer, Berlin, Heidelberg.
- [11] Czeskis, A., Dermendjieva, I., Yapit, H., Borning, A., Friedman, B., Gill, B., & Kohno, T. (2010, July). Parenting from the pocket: Value tensions and technical directions for secure and private parent-teen mobile safety. In *Proceedings of the sixth symposium on usable privacy and security* (pp. 1-15). <https://doi.org/10.1145/1837110.1837130>
- [12] Dourish, P. (2006). Implications for design. In *Proceedings of the SIGCHI conference on Human Factors in computing systems* (pp. 541-550). <https://doi.org/10.1145/1124772.1124855>
- [13] Felt, U. (2017). "Response-able practices" or "new bureaucracies of virtue": the challenges of making RRI work in academic environments. In *Responsible innovation 3* (pp. 49-68). Springer, Cham.



- [14] Fisher, E., Mahajan, R. L., & Mitcham, C. (2006). Midstream modulation of technology: governance from within. *Bulletin of Science, Technology & Society*, 26(6), 485-496. <https://doi.org/10.1177/0270467606295402>
- [15] Friedman, B., Hendry, D. G., & Borning, A. (2017). A survey of value sensitive design methods. *Foundations and Trends® in Human-Computer Interaction*, 11(2), 63-125. <http://dx.doi.org/10.1561/11000000015>
- [16] Friedman, B., Kahn Jr, P. H., Hagman, J., Severson, R. L., & Gill, B. (2006). The watcher and the watched: Social judgments about privacy in a public place. *Human-Computer Interaction*, 21(2), 235-272. https://doi.org/10.1207/s15327051hci2102_3
- [17] Friedman, B., Kahn, P. H., Borning, A., & Hultgren, A. (2013). Value sensitive design and information systems. In *Early engagement and new technologies: Opening up the laboratory* (pp. 55-95). Springer, Dordrecht.
- [18] Government Office for Science (2017). The Futures Toolkit. Tools for Futures Thinking and Foresight Across UK Government. Retrieved December 2, 2022, from https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/674209/futures-toolkit-edition-1.pdf
- [19] Hagendorff, T. (2020). The ethics of AI ethics: An evaluation of guidelines. *Minds and Machines*, 30(1), 99-120. <https://doi.org/10.1007/s11023-020-09517-8>
- [20] Hine, C. (2008). Virtual ethnography: Modes, varieties, affordances. *The SAGE handbook of online research methods*, 257-270. Sage: London.
- [21] Ideo (2015). The Field Guide to Human-Centered Design. Retrieved December 2, 2022, from https://d1r3w4d5z5a88i.cloudfront.net/assets/guide/Field%20Guide%20to%20Human-Centered%20Design_IDEOorg_English-0f60d33bce6b870e7d80f9cc1642c8e7.pdf
- [22] Institute for the Future (2018). Ethical Operating System. Retrieved December 2, 2022, from <https://ethicalos.org/>
- [23] Jobin, A., Ienca, M., & Vayena, E. (2019). The global landscape of AI ethics guidelines. *Nature Machine Intelligence*, 1(9), 389-399. <https://doi-org/10.1038/s42256-019-0088-2>
- [24] Lelea, M. A., Roba, G. M., Christinck, A., & Kaufmann, B. (2014) Methodologies for Stakeholder Analysis. Retrieved December 2, 2022, from [http://reload-globe.net/cms/attachments/article/56/Lelea_et_al_\(2014\)_StakeholderGuide_final_web.pdf](http://reload-globe.net/cms/attachments/article/56/Lelea_et_al_(2014)_StakeholderGuide_final_web.pdf)
- [25] Living Innovation (2019). Co-Creation Toolkit. Retrieved December 2, 2022, from <https://www.living-innovation.net/news/article?id=212&title=new-toolkit-for-effective-co-creation>
- [26] Longo, F., Padovano, A., & Umbrello, S. (2020). Value-oriented and ethical technology engineering in industry 5.0: A human-centric perspective for the design of the factory of the future. *Applied Sciences*, 10(12), 4182. <http://dx.doi.org/10.3390/app10124182>
- [27] McMillan, D., & Brown, B. (2019). Against ethical AI. In *Proceedings of the Halfway to the Future Symposium 2019* (pp. 1-3). <https://doi.org/10.1145/3363384.3363393>



- [28] Miller, J. K., Friedman, B., Jancke, G., & Gill, B. (2007). Value tensions in design: the value sensitive design, development, and appropriation of a corporation's groupware system. In *Proceedings of the 2007 international ACM conference on Supporting group work* (pp. 281-290). <https://doi.org/10.1145/1316624.1316668>
- [29] Nathan et al. (2005). Value scenarios draw upon 5 key elements in order to develop provocative sketchers of the future: stakeholders, pervasiveness, time, systemic effect, and value implications
- [30] Nathan, L. P., Friedman, B., Klasnja, P., Kane, S. K., & Miller, J. K. (2008). Envisioning systemic effects on persons and society throughout interactive system design. In *Proceedings of the 7th ACM conference on Designing interactive systems* (pp. 1-10). <https://doi.org/10.1145/1394445.1394446>
- [31] NewHoRRizon (2018). Ensuring Societal Readiness: A Thinking Tool. Retrieved December 2, 2022, from https://thinkingtool.eu/Deliverable_6.1_Final_April%2030_THINKING_TOOL.pdf
- [32] Patrick, G. (2021). Stakeholder Mapping: Identify & Assess Project Stakeholders. Boréal, Retrieved December 2, 2022, from www.boreal-is.com/blog/stakeholder-mapping-identify-stakeholders
- [33] Reeves, S., Kuper, A., & Hodges, B. D. (2008). Qualitative research methodologies: ethnography. 337. <https://doi.org/10.1136/bmj.a1020>
- [34] Rip, A., & Kemp, R. (1998). Technological change. *Human choice and climate change*, 2(2), 327-399.
- [35] Roulston, K., & Choi, M. (2018). Qualitative interviews. *The SAGE handbook of qualitative data collection*, 233-249. Sage: London.
- [36] Sanders, E. B. N., & Westerlund, B. (2011). Experiencing, exploring and experimenting in and with co-design spaces. *Nordes*, (4). Retrieved December 2, 2022, from <https://archive.nordes.org/index.php/n13/article/view/110>
- [37] Schuurbiers, D., & Fisher, E. (2009). Lab-scale intervention. *EMBO reports*, 10(5), 424-427. <https://doi.org/10.1038/embor.2009.80>
- [38] Shilton, K. (2013). Values levers: Building ethics into design. *Science, Technology, & Human Values*, 38(3), 374-397. <https://doi.org/10.1177/0162243912436985>
- [39] Shriram, K., & Schwartz, R. (2017). All are welcome: Using VR ethnography to explore harassment behavior in immersive social virtual reality. IEEE. <https://doi.org/10.1109/VR.2017.7892258>
- [40] Stevanovic, M., Marjanovic, D., & Storga, M. (2015). A model of idea evaluation and selection for product innovation. In *DS 80-8 Proceedings of the 20th International Conference on Engineering Design (ICED 15) Vol 8: Innovation and Creativity*, Milan, Italy, 27-30.07. 15 (pp. 193-202). Retrieved December 2, 2022, from <https://www.designsociety.org/publication/37913/A+MODEL+OF+IDEA+EVALUATION+AND+SELECTION+FOR+PRODUCT+INNOVATION>
- [41] Stilgoe, Jack, Richard Owen, and Phil Macnaghten. "Developing a Framework for Responsible Innovation." *Research Policy* 42, no. 9 (2013): 1568-1580. <https://doi.org/10.1016/j.respol.2013.05.008>

- [42] van de Poel, I. (2020). Embedding values in artificial intelligence (AI) systems. *Minds and Machines*, 30(3), 385-409. <https://doi.org/10.1007/s11023-020-09537-4>
- [43] Wagner, B. (2018). Ethics as an escape from regulation. From “ethics-washing” to ethics-shopping? *Legal and Political Theory in Data-Driven Environments*. Retrieved December 2, 2022, from https://mediarep.org/bitstream/handle/doc/14203/Being-profiled_84-88_Wagner_Ethics_as_an_escape.pdf?sequence=1
- [44] Yoo, D. (2018). Stakeholder Tokens: A constructive method for value sensitive design stakeholder analysis. *Ethics and Information Technology*. <https://doi.org/10.1007/s10676-018-9474-4>.
- [45] Yoo, D., Hultgren, A., Woelfer, J. P., Hendry, D. G., & Friedman, B. (2013, April). A value sensitive action-reflection model: evolving a co-design space with stakeholder and designer prompts. In *Proceedings of the SIGCHI conference on human factors in computing systems* (pp. 419-428). <https://doi.org/10.1145/2470654.2470715>